

Electromagnetic Induction, solved

Step-by-step solutions to every practice problem in Chapter 20. Try each problem on your own before you read its solution.

By **Pratik Bhanuse**, PhD Physics candidate, Western University ·
pratikbhanuse.github.io

P20.1

Copper pipe. Copper is a good conductor. As the magnet falls, the flux through the rings of pipe ahead of it increases, and the flux through the rings behind it decreases. Both changes induce eddy currents. By Lenz's law, the currents oppose the change: the ring ahead repels the magnet, and the ring behind attracts it back. The magnet soon reaches a slow terminal speed at which these magnetic forces balance its weight.

Plastic pipe. Plastic is an insulator, so no currents can flow. The magnet falls freely.

Energy. At terminal speed, the magnet's lost gravitational energy is dissipated as **heat** in the copper, through the resistance met by the eddy currents. The pipe warms very slightly.

P20.2

(a) **Yes.** The changing field produces an induced electric field, $\oint \vec{E} \cdot d\vec{l} = -d\Phi/dt$, which pushes the charges round the loop. This is an **electric** force.

(b) **No**, as long as the loop stays entirely within the uniform field and does not rotate. The flux through it is constant. In terms of forces, the $\vec{v} \times \vec{B}$ EMFs in opposite sides of the loop are equal and cancel.

(c) **Yes.** As the loop enters the field, the flux through it increases. Here the charges are driven by the **magnetic** force $q\vec{v} \times \vec{B}$ on the moving charges in the part of the loop that is inside the field, which gives $\mathcal{E} = BLv$.

Faraday's law covers both mechanisms in a single statement. Einstein's realisation that these are the same phenomenon, seen from different frames, was a starting point for special relativity.

P20.3

The alternating flux in the core induces eddy currents within the iron itself, which waste energy as heat and can make the core very hot. Splitting the core into thin sheets, insulated from each other and lying parallel to the flux, cuts the large current loops into narrow strips of much higher resistance. Eddy-current loss scales roughly as the *square* of the sheet thickness, so laminations about 0.3 mm thick cut the losses enormously. At high frequencies, ferrite (an insulating magnetic ceramic) or powdered iron is used instead.

P20.4

An electromagnet has a large inductance, and it carries a steady current. Opening the switch tries to stop that current almost instantly, so dI/dt is huge, and the induced EMF $L dI/dt$ can reach thousands of volts. Lenz's law means this EMF tries to *keep the current flowing*. The voltage appears across the opening switch contacts and ionises the air between them, producing a spark through which the current briefly continues. The magnetic energy $\frac{1}{2}LI^2$ is dumped into the spark, which erodes the contacts.

P20.5

$$|\mathcal{E}| = NA \left| \frac{dB}{dt} \right| = 50(0.10)(0.20) = 1.0 \text{ V}, \quad I = \frac{1.0}{5.0} = 0.20 \text{ A}.$$

Direction. The flux is *decreasing*, so the induced current produces a field **in the same direction** as the applied field, trying to maintain the flux. Looking along $\vec{B}$ (with the field pointing away from you), the current circulates **clockwise**, by the right-hand grip rule.

P20.6

$$\omega = 2\pi(60) = 377 \text{ rad/s}.$$

$$\mathcal{E}_0 = NBA\omega = 50(0.20)(0.010)(377) = 38 \text{ V}, \quad \mathcal{E}(t) = 38 \sin(377t) \text{ V}.$$

(Taking $t = 0$ when the plane of the coil is perpendicular to $\vec{B}$.)

P20.7

Terminal velocity. The magnetic drag on the rod is $B^2 L^2 v / R$, and it grows with speed. At terminal velocity it balances the applied force:

$$F = \frac{B^2 L^2 v_t}{R} \Rightarrow v_t = \frac{FR}{B^2 L^2} = \frac{0.50(2.0)}{(0.50)^2(0.40)^2} = \frac{1.0}{0.040} = 25 \text{ m/s.}$$

Power: $P = Fv_t = 0.50(25) = 12.5 \text{ W.}$

Check: $I = BLv_t/R = (0.50)(0.40)(25)/2.0 = 2.5 \text{ A}$, so $I^2 R = 12.5 \text{ W.}$ ✓ All the work done by the force becomes heat in the resistance.

P20.8

$$\mathcal{E} = BLv = (4.0 \times 10^{-5})(60)(250) = 0.60 \text{ V.}$$

Not useful. To draw current, a complete circuit is needed. Any wire run from wingtip to wingtip *inside* the aircraft moves with it through the same field, so it has exactly the same induced EMF, which cancels the first. No current flows. The wing simply acquires a slight charge separation, with one tip slightly positive and the other slightly negative.

P20.9

$$A = \pi(0.020)^2 = 1.26 \times 10^{-3} \text{ m}^2.$$

$$L = \frac{\mu_0 N^2 A}{\ell} = \frac{(4\pi \times 10^{-7})(500)^2(1.26 \times 10^{-3})}{0.25} = 1.6 \times 10^{-3} \text{ H.}$$

$$U = \frac{1}{2} LI^2 = \frac{1}{2} (1.58 \times 10^{-3})(3.0)^2 = 7.1 \times 10^{-3} \text{ J.}$$

P20.10

$$V_2 = V_1 \frac{N_2}{N_1} = 230 \times \frac{60}{1150} = 12 \text{ V.}$$

$$I_2 = \frac{V_2}{R} = \frac{12}{4.0} = 3.0 \text{ A,} \quad I_1 = I_2 \frac{N_2}{N_1} = 3.0 \times \frac{60}{1150} = 0.16 \text{ A.}$$

$$P = V_2 I_2 = 36 \text{ W,} \quad V_1 I_1 = 230(0.157) = 36 \text{ W.} \checkmark$$

P20.11

$$\tau = \frac{L}{R} = \frac{2.0}{50} = 0.040 \text{ s} = 40 \text{ ms.}$$

The current reaches 90% when $e^{-t/\tau} = 0.10$:

$$t = \tau \ln 10 = 40 \text{ ms} \times 2.30 = 92 \text{ ms.}$$

P20.12

The flux through each turn is $\Phi = AB_0 \sin \omega t$, so the peak EMF is

$$\mathcal{E}_0 = NAB_0\omega = 30 \times \pi(0.010)^2 \times (50 \times 10^{-6}) \times 2\pi(1.0 \times 10^5).$$

$$\mathcal{E}_0 = 30 \times 3.14 \times 10^{-4} \times 5.0 \times 10^{-5} \times 6.28 \times 10^5 = 0.30 \text{ V.}$$

After rectification and voltage boosting, this is enough to run or recharge low-power implant electronics. Real systems tune both coils to resonance (Chapter 21) to transfer power more efficiently.

P20.13

(a) **Terminal speed.** Only the upper edge, of length L , is inside the field. As the loop falls with speed v , the area of loop inside the field shrinks at the rate Lv , so

$$\mathcal{E} = BLv, \quad I = \frac{BLv}{R}.$$

The current in the upper edge feels a magnetic force $ILB = B^2L^2v/R$. By Lenz's law it points **upward**, opposing the fall. (The forces on the two vertical sides cancel.) Terminal speed is reached when this force balances the weight:

$$mg = \frac{B^2L^2v_t}{R} \Rightarrow v_t = \frac{mgR}{B^2L^2}.$$

(b) **Numbers:**

$$v_t = \frac{(0.010)(9.8)(0.050)}{(1.0)^2(0.10)^2} = 0.49 \text{ m/s.}$$

(c) **Energy.** At terminal speed, the kinetic energy is constant, so gravitational energy is lost at the rate mgv_t . It all appears as Joule heat in the loop: $I^2R = (BLv_t)^2/R = mgv_t$. ✓ The same

principle is used in magnetic brakes for drop-tower amusement rides.

P20.14

(a) **Current I_1 in the inner solenoid.** It produces $B_1 = \mu_0 N_1 I_1 / \ell$ only inside its own cross-section A_1 , and nothing outside it. The flux through each turn of the outer coil is therefore $B_1 A_1$, and the flux linkage is

$$N_2 B_1 A_1 = \frac{\mu_0 N_1 N_2 A_1}{\ell} I_1 \Rightarrow M_{21} = \frac{\mu_0 N_1 N_2 A_1}{\ell}.$$

Current I_2 in the outer solenoid. It produces $B_2 = \mu_0 N_2 I_2 / \ell$ everywhere inside the outer coil, including inside the inner one. The flux linkage of the inner coil is

$$N_1 B_2 A_1 = \frac{\mu_0 N_1 N_2 A_1}{\ell} I_2 \Rightarrow M_{12} = \frac{\mu_0 N_1 N_2 A_1}{\ell} = M_{21}. \checkmark$$

(b) **Numbers:**

$$M = \frac{(4\pi \times 10^{-7})(400)(800)(2.0 \times 10^{-4})}{0.20} = 4.0 \times 10^{-4} \text{ H} = 0.40 \text{ mH}.$$

P20.15

With $L = \mu_0 N^2 A / \ell$ and $B = \mu_0 N I / \ell$:

$$\frac{1}{2} L I^2 = \frac{1}{2} \frac{\mu_0 N^2 A}{\ell} I^2 = \frac{1}{2\mu_0} \left(\frac{\mu_0 N I}{\ell} \right)^2 A \ell = \frac{B^2}{2\mu_0} \times (A \ell).$$

$A \ell$ is the volume inside the solenoid, where the uniform field is, and the field outside is negligible. So the integral of $u_B = B^2 / 2\mu_0$ over all space equals $\frac{1}{2} L I^2$. $\checkmark$

Interpretation. The energy put in to establish the current is not stored “in the wire”. It is stored in the **magnetic field** filling the space inside the coil, just as a capacitor’s energy is stored in its electric field. When the current is switched off, this field energy is returned to the circuit.

P20.16

(a) **Induced field.** By symmetry, the induced field lines are circles about the axis, and E is the same all the way round each circle.

- Inside ($r < R$):

$$E(2\pi r) = \pi r^2 \frac{dB}{dt} \Rightarrow E = \frac{r}{2} \frac{dB}{dt}.$$

- Outside ($r > R$):

$$E(2\pi r) = \pi R^2 \frac{dB}{dt} \Rightarrow E = \frac{R^2}{2r} \frac{dB}{dt}.$$

The field is largest at $r = R$.

(b) Torso:

$$E_{\max} = \frac{R}{2} \frac{dB}{dt} = \frac{0.20}{2} (50) = 5.0 \text{ V/m}.$$

This is comparable to the thresholds for stimulating peripheral nerves. Scanners therefore limit the gradient switching rate (and the product of amplitude and slew rate) to keep dB/dt below the levels that cause twitching or discomfort. They monitor this in real time, and patients are warned that they may feel tingling during fast imaging sequences. The limits are tighter for cardiac stimulation, which would be dangerous, but those thresholds are far higher.

P20.17

(a) **EMF.** A radial element of the disc between r and $r + dr$ moves at speed ωr perpendicular to $\vec{B}$, so it contributes a motional EMF $B(\omega r) dr$. All these elements are in series along the radius:

$$\mathcal{E} = \int_0^R B\omega r dr = \frac{1}{2} B\omega R^2.$$

(b) **Numbers.** $\omega = 3000 \times 2\pi/60 = 314 \text{ rad/s}$:

$$\mathcal{E} = \frac{1}{2} (1.0)(314)(0.50)^2 = 39 \text{ V}.$$

(c) **Large currents, low voltage.** A disc is effectively a *single turn*, so the EMF is modest: you cannot stack up turns as in a coil. But the solid copper path from axle to rim has a tiny resistance, so extremely large currents (up to millions of amperes in pulsed machines) can be drawn. Homopolar generators and motors are used for pulsed power, railgun research and some marine propulsion systems.

P20.18

(a) **Steady state:**

$$I = \frac{12}{24} = 0.50 \text{ A}, \quad U = \frac{1}{2}LI^2 = \frac{1}{2}(10)(0.50)^2 = 1.25 \text{ J}.$$

(b) **Interrupting the current:**

$$|\mathcal{E}| \approx L \frac{\Delta I}{\Delta t} = 10 \times \frac{0.50}{1.0 \times 10^{-3}} = 5.0 \times 10^3 \text{ V}.$$

Five thousand volts would arc across the switch or destroy a transistor driving the relay.

(c) **With a flyback diode.** The current now decays through the coil's own resistance:

$$\tau = \frac{L}{R} = \frac{10}{24} = 0.42 \text{ s}.$$

When the switch opens, the inductor keeps driving its current, but now round the loop formed by the diode and the coil. The voltage across the coil is limited to about the diode's forward voltage (roughly 0.7 V), and the switch sees only about the supply voltage. The stored 1.25 J is dissipated gently as heat in the coil over a fraction of a second. The trade-off is that the relay releases a little more slowly.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

© 2026 Pratik Bhanuse. All rights reserved. · pratikbhanuse.github.io